SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	VINIL KUMAR S	**Reg. No.:**	192511226
Section / Batch:		**Date:**	24/07/2026

Code of Conduct

I VINIL KUMAR S (192511226) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: VINIL KUMAR S

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
	GRAND TOTAL:	**100 Marks**	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error:

```
if((ch == ')' && stack[top] == '(') ||
(ch == ']' && stack[top] == '[')) {
pop();
} else {
return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

char stack[100];
int top = -1;

void push(char ch) {
    stack[++top] = ch;
}

void pop() {
    top--;
}

int main() {
    char ch;
    int balanced = 1;

    printf("enter brackets: ");

    while (scanf("%c", &ch) == 1 && ch != '\n') {

        if (ch == '(' || ch == '[' || ch == '{') {
            push(ch);
        }
        else if (ch == ')' || ch == ']' || ch == '}') {

            if (top == -1) {
                balanced = 0;
                break;
            }

            if ((ch == ')' && stack[top] == '(') ||
                (ch == ']' && stack[top] == '[') ||
                (ch == '}' && stack[top] == '{')) {
                pop();
            }
            else {
                balanced = 0;
                break;
            }
        }
    }

    if (top != -1) {
        balanced = 0;
    }

    if (balanced) {
        printf("balanced brackets\n");
    }
    else {
        printf("not balanced brackets\n");
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
enter brackets: balanced brackets
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q10 **Queue** 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 100

int queue[MAX];
int front = -1;
int rear = -1;

int dequeue() {
    int item;

    if (front == -1) {
        printf("Underflow\n");
        return -1;
    }

    item = queue[front];
    front++;

    if (front > rear) {
        front = -1;
        rear = -1;
    }

    return item;
}

int main() {
    int value;

    // Adding elements
    queue[++rear] = 10;
    if (front == -1)
        front = 0;

    queue[++rear] = 20;
    queue[++rear] = 30;

    value = dequeue();

    if (value != -1) {
        printf("Dequeued element: %d\n", value);
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Dequeued element: 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

Q9 Linked List

5 marks

What is the error in the below code?

new.next = head

head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

int main() {
    struct Node *head = NULL;
    struct Node *newNode;

    newNode = (struct Node *)malloc(sizeof(struct Node));

    newNode->data = 10;
    newNode->next = head;
    head = newNode;

    printf("Node inserted successfully\n");
    printf("Data: %d\n", head->data);

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Node inserted successfully
Data: 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Q8 **Linked List** 5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

int main() {
    struct Node *head, *current, *prev, *next;

    head = (struct Node *)malloc(sizeof(struct Node));
    head->data = 10;

    head->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->data = 20;

    head->next->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->next->data = 30;

    head->next->next->next = NULL;

    current = head;
    prev = NULL;

    while (current != NULL) {
        next = current->next;
        current->next = prev;
        prev = current;
        current = next;
    }

    head = prev;

    current = head;

    while (current != NULL) {
        printf("%d ", current->data);
        current = current->next;
    }

    return 0;
}
```

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Test Input

stdin input (optional)

Output

```
30 20 10
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 4 marks
- Q6 Single Linked List 4 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
int binarysearch(int arr[], int n, int key)
{
    int low = 0, high=n-1;
    while(low<=high)
    {
        int mid=low+(high-low)/2;
        if(arr[mid]==key)
            return mid;
        else if (arr[mid]<key)
            low=mid+1;
        else
            high=mid-1;
    }
    return -1;
}
int main() {
    int n,key;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
        scanf("%d",&arr[i]);
    scanf("%d",&key);
    int result=binarysearch(arr,n,key);
    printf("%d",result);
    return 0;
}
```

Test Input

```
5
2 5 8 12 16
12
```

Output

```
[illegible]
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q7 **Stack** 5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 100

int stack[MAX];
int top = -1;

int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }
    else {
        return stack[top--];
    }
}

int main() {
    int value;

    stack[++top] = 10;
    stack[++top] = 20;
    stack[++top] = 30;

    value = pop();

    if (value != -1) {
        printf("Popped element: %d\n", value);
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Popped element: 30
```

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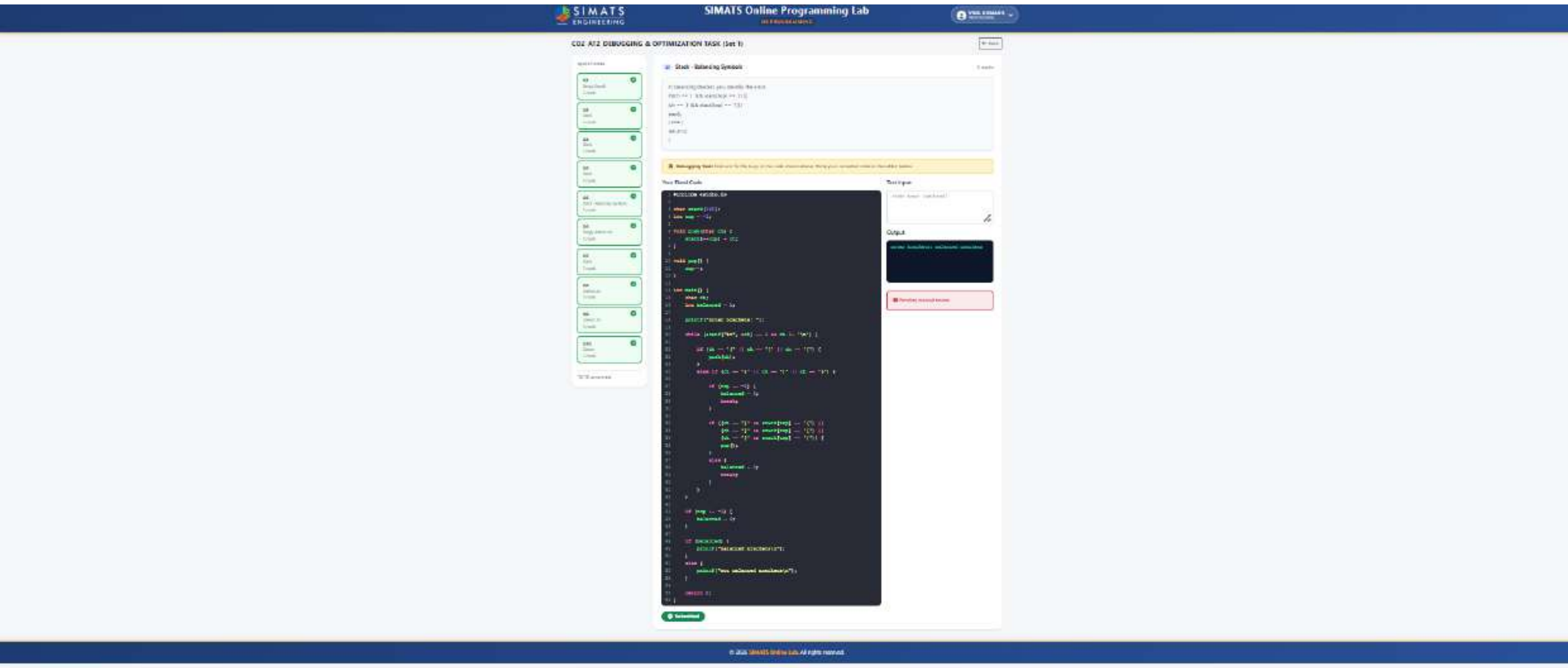
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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 1)
Stack - Balancing Symbols
#include <stdio.h>
Output
Submitted
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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1
- Q2 Stack
- Q3 Stack
- Q4 Stack
- Q5 Stack - Balancing Symbols
- Q6
- Q7 Stack
- Q8
- Q9
- Q10

10/10 answered

Q4 Stack — 5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != '(') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int main() {
    char ch;

    printf("Enter a character: ");
    scanf(" %c", &ch);

    if (ch == ')') {
        printf("Closing parenthesis\n");
    } else {
        printf("Not a closing parenthesis\n");
    }

    return 0;
}
```

Test Input

```
)
```

Output

```
Enter a character: Closing
parenthesis
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1
- Q2
- Q3
- Q4
- Q5
- Q6
- Q7
- Q8
- Q9
- Q10

10/10 answered

Q4 **Stack**

5 marks

Check the snippet and identify the logical error.
int stack[5];
int top;
void init() {
top = 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include<stdio.h>
int stack[5];
int top;
void init() {
    top=-1;
}
int main() {
    init();
    printf("top=%d\n",top);
    return 0;
}
```

Test Input

stdin input (optional)

Output

Run | Save | Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search
- Q2 Stack
- Q3 Stack
- Q4 Stack
- Q5 Stack - Balancing symbol
- Q6 Singly Linked List
- Q7 Stack
- Q8 Linked List
- Q9 Linked List
- Q10 Queue

10/10 answered

Q2 Stack — 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include<stdio.h>
int stack[100];
int top=2;
int pop(){
    if(top==-1){
        printf("underflow");
        return -1;
    }
    return stack[top--];
}
int main(){
    stack[0]=10;
    stack[1]=20;
    stack[2]=30;
    printf("%d\n",pop());
    return 0;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q1 **Stack** 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 100

int stack[MAX];
int top = -1;

int peek() {
    if (top == -1) {
        printf("Stack Underflow\n");
        return -1;
    }

    return stack[top];
}

int main() {
    stack[++top] = 10;
    stack[++top] = 20;
    stack[++top] = 30;

    printf("Top element: %d\n", peek());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Top element: 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int vehicle) {

    if ((rear + 1) % MAX == front) {
        printf("Queue Full\n");
        return;
    }

    if (front == -1) {
        front = 0;
        rear = 0;
    }
    else {
        rear = (rear + 1) % MAX;
    }

    queue[rear] = vehicle;

    printf("Vehicle %d added\n", vehicle);
}

int main() {
    enqueue(101);
    enqueue(102);
    enqueue(103);
    enqueue(104);
    enqueue(105);
    enqueue(106);

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Vehicle 101 added
Vehicle 102 added
Vehicle 103 added
Vehicle 104 added
Vehicle 105 added
Queue Full
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q3 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

void push(int value) {
    if (top == MAX - 1) {
        printf("Overflow\n");
        return;
    }

    stack[++top] = value;
    printf("Pushed: %d\n", value);
}

int main() {
    push(10);
    push(20);
    push(30);
    push(40);
    push(50);
    push(60);

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Pushed: 10
Pushed: 20
Pushed: 30
Pushed: 40
Pushed: 50
Overflow
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }

    return stack[top--];
}

int main() {
    stack[++top] = 10;
    stack[++top] = 20;
    stack[++top] = 30;

    printf("Popped element: %d\n", pop());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Popped element: 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *prev;
    struct Node *next;
};

void deleteNode(struct Node **head, struct Node *del) {

    if (del == NULL || *head == NULL)
        return;

    if (del->prev != NULL)
        del->prev->next = del->next;
    else
        *head = del->next;

    if (del->next != NULL)
        del->next->prev = del->prev;

    free(del);
}

int main() {
    struct Node *head = NULL;
    struct Node *del;

    head = (struct Node *)malloc(sizeof(struct Node));
    head->data = 10;
    head->prev = NULL;

    head->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->data = 20;
    head->next->prev = head;
    head->next->next = NULL;

    del = head->next;

    deleteNode(&head, del);

    printf("Remaining node: %d\n", head->data);

    return 0;
}
```

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Test Input

```
stdin input (optional)
```

Output

```
Remaining node: 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1 Stack 5 marks
Q2 Circular Queue 5 marks
Q3 Stack 5 marks
Q4 Stack 5 marks
Q5 Doubly Linked List 5 marks
Q6 Queue 5 marks
Q7 Queue · Deque 5 marks
Q8 Binary Search 5 marks
Q9 Circular Queue 5 marks
Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q6 **Queue** 5 marks

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 5

int queue[MAX];
int rear = 0;

void enqueue(int data) {
    if (rear == MAX) {
        printf("Queue Overflow\n");
        return;
    }

    queue[rear++] = data;
}

int main() {
    enqueue(10);
    enqueue(20);
    enqueue(30);
    enqueue(40);
    enqueue(50);
    enqueue(60);

    printf("Queue elements: ");

    for (int i = 0; i < rear; i++) {
        printf("%d ", queue[i]);
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Queue Overflow
Queue elements: 10 20 30 40 50
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q7 **Queue - Deque** 5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int deque[MAX];
int front = -1;
int rear = -1;

void insertFront(int data) {

    if ((front == 0 && rear == MAX - 1) ||
        (front == rear + 1)) {
        printf("Deque Overflow\n");
        return;
    }

    if (front == -1) {
        front = 0;
        rear = 0;
    }
    else if (front == 0) {
        front = MAX - 1;
    }
    else {
        front = front - 1;
    }

    deque[front] = data;
}

int main() {
    insertFront(10);
    insertFront(20);
    insertFront(30);

    printf("Deque elements: ");

    int i = front;

    while (1) {
        printf("%d ", deque[i]);

        if (i == rear)
            break;

        i = (i + 1) % MAX;
    }

    return 0;
}
```

Submitted

Test Input

stdin input (optional)

Output

```
Deque elements: 30 20 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 — Stack — 5 marks
- Q2 — Circular Queue — 5 marks
- Q3 — Stack — 5 marks
- Q4 — Stack — 5 marks
- Q5 — Doubly Linked List — 5 marks
- Q6 — Queue — 5 marks
- Q7 — Queue - Deque — 5 marks
- Q8 — Binary Search — 5 marks
- Q9 — Circular Queue — 5 marks
- Q10 — Stack - Postfix Expression — 5 marks

10/10 answered

Q8 **Binary Search** 5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int binarySearch(int arr[], int l, int r, int x) {

    if (l > r)
        return -1;

    int mid = l + (r - l) / 2;

    if (arr[mid] == x)
        return mid;

    if (arr[mid] > x)
        return binarySearch(arr, l, mid - 1, x);

    return binarySearch(arr, mid + 1, r, x);
}

int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int n = 5;
    int x = 30;

    int result = binarySearch(arr, 0, n - 1, x);

    if (result != -1)
        printf("Element found at index %d\n", result);
    else
        printf("Element not found\n");

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Element found at index 2
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Dequeue 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

int dequeue() {
    int data;

    if (front == -1) {
        printf("Queue Underflow\n");
        return -1;
    }

    data = queue[front];

    if (front == rear) {
        front = -1;
        rear = -1;
    }
    else {
        front = (front + 1) % MAX;
    }

    return data;
}

int main() {
    queue[0] = 10;
    queue[1] = 20;
    queue[2] = 30;

    front = 0;
    rear = 2;

    printf("Dequeued: %d\n", dequeue());
    printf("Dequeued: %d\n", dequeue());
    printf("Dequeued: %d\n", dequeue());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Dequeued: 10
Dequeued: 20
Dequeued: 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Dequeue
5 marks

Q10 **Stack - Postfix Expression**

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

int main() {
    char expression[100];
    int stack[100];
    int top = -1;
    int i;
```

Test Input

stdin input (optional)

Output